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No Evidence for the Efficacy of Slow-Paced Breathing as a Recovery Strategy After Sprint Interval Training

in [International Journal of Sports Physiology and Performance](#)

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Peter Raidl *, Barbara Wessner, Simon Laister, and Robert Csapo

DOI: <https://doi.org/10.1123/ijsp.2025-0138>

Keywords: [breathing exercise](#); HRV; HIIT

First Published Online: 12 Jan 2026

In Print: [Volume 21: Issue 2](#)

Page Range: 302–311

Purpose: Breathing practices are increasingly adopted by athletes due to the potential benefits for recovery, performance, and well-being. Slow-paced breathing is believed to increase cardiac vagal activity and relaxation, potentially supporting adaptive processes. Sprint interval training, by contrast, elicits a strong stress response. Therefore, we hypothesized that adding slow-paced breathing to highly demanding sprint interval training would enhance recovery, as indicated by cardiac autonomic control and sleep metrics, ultimately contributing to greater performance gains. **Methods:** Thirty-four participants were randomized into an intervention or control group. While both groups completed 4 weeks of supervised sprint interval training, only the intervention group practiced slow-paced breathing after each training and every evening before bedtime. Running performance, vagally mediated heart-rate variability (root mean square of successive differences), and sleep metrics were assessed throughout the intervention to evaluate the potential benefits of the breathing exercise. **Results:** Sprint interval training led to significant improvements in maximal power decrements (-8.26% , 95% CI, -11.48 to -5.04 , $P < .001$) and overall work (8.84 kJ/kg, 95% CI, 4.58 to 13.1 , $P < .001$), but performance gains did not differ between groups ($P > .05$). Unexpectedly, nocturnal heart rate showed a slight increase over time in the intervention group ($+3.1$ beats·min $^{-1}$, 95% CI, 0.47 to 5.72 , $P < .05$) compared with the control group. **Conclusion:** Despite positive effects reported in other training and performance contexts, 4 weeks of daily slow-paced breathing did not confer additional performance improvements or physiological benefits when combined with sprint interval training. These findings suggest that 10-minute sessions of slow-paced breathing following highly demanding sprint interval training offer limited value for enhancing performance or supporting recovery.